CORPORATE CULTURE IN THE BANKING SECTOR:

Measuring a Key Intangible at BBVA

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TABLE OF CONTENTS

CORPORATE CULTURE IN THE BANKING SECTOR:.....1
1. Summary.....4
2. Introduction.....5
3. Research Question and Objectives.....5
3.1 Main Research Question and Objectives.....5
4. Literature Review.....6
4.1 Defining Corporate Culture.....6
4.2 Culture and Employee Outcomes.....6
4.3 Culture and Financial Performance.....6
5. Conceptual Framework.....7
5.1 BBVA Values and Expected Behaviors.....7
5.2 Value-to-Proxy Mapping.....7
5.3 Conceptual Model.....7
6. BBVA Corporate Culture and Values.....8
6.1 Current Value System.....8
6.2 Current Measurement Practices.....8
6.3 Identified Perception Gaps.....8
7. Methodology.....9
7.1 How We Arrived at This Approach.....9
7.2 Index Construction Logic.....9
7.3 Variable Selection.....10
7.4 Standardization.....12
7.5 Aggregation.....12
7.6 Alternative Weighting Schemes.....12
7.7 Worked Calculations.....12
7.7.1 Worked Example A. Employee-Level Illustration (Demonstration Values).....12
7.7.2 Worked Example B. Bank-Level Computation with Observed Refinitiv LSEG Data.....13
7.8 Econometric Models.....14
Model 1: Does stronger culture keep people from leaving?.....14
Model 2: Does culture show up in satisfaction?.....14
Model 3: Does culture spark ideas?.....15
Model 4: Do banks with stronger culture perform better?.....15
Model 5 : Do Spanish regions differ? (In case the bank has regional data).....15
Summary of Hypotheses.....16
7.9 Expected Relationships.....16
8. Data Description and Strategy.....17
8.1 Overview.....17
8.2 Survey Data.....17
8.3 External Public Data.....17
8.4 Artificial Data Generation: Three-Scenario Replication.....18
8.5 Indicator Sourcing Without Internal Data.....18
8.6 The Synthetic Employee Data and How It Behaves.....18

8.7 Proposed Dataset Structure....20
8.7.1 Hierarchical Design....20
8.7.2 Employee-Year Dataset (Artificially Generated and Replicated)....20
8.7.3 Department-Year and Bank-Year Datasets....20
8.7.4 Pilot Implementation of the Dataset....21
8.8 Observed Data: the Refinitiv Extraction (real_data.xlsx)....21
9. Implementation and Results....22
9.1 The Pilot Analysis Workbook (BBVA_Dataset.xlsx)....22
9.1.1 Formula Breakdown....23
9.2 How to Use the Dataset, and Why It Is Adjustable....24
9.3 Bank-Level Pilot Analysis Results....24
9.4 Interpretation of Results....27
9.4.1 What the pilot results mean....27
9.4.2 What the results do not mean....27
9.4.3 How to interpret once real survey data arrive....27
10. Practical Implementation....28
11. Scope and Limitations....29
11.1 Scope....29
11.2 Limitations....29
11.3 Theoretical and Methodological Limitations....29
12. Further Steps....30
12.1 Validation and Robustness Checks....30
12.2 How the Limitations Are Advised to Be Addressed....31
References....32
Annexes....33
Annex A. Variable Dictionary (pilot)....33
Annex B. Observed Benchmark Data (LSEG Workspace extraction)....33
Annex C. Culture Perception Survey (instrument underlying the employee replication)....33
Section A . About you (anonymous, categorical bands)....34
Section B . Satisfaction and Well-being....34
Section C . Ethics and Integrity....34
Section D . Engagement, Psychological Safety, and Teamwork (“We Are One Team”)....34
Section E . Leadership Trust....34
Section F . Growth and Innovation (“We Think Big”)....35
Section G . Customer Orientation (“Customer Comes First”)....35
Section H . Value Implementation....35
Scoring and Mapping to the Dataset....35

1. Summary

Corporate culture is one of the most important intangible assets that influences organizational performance, employee behavior, customer trust, and long-term competitiveness of the company. Despite its strategic relevance, culture remains difficult to measure due to its intangible nature and the challenge of translating organizational values into observable and quantifiable behaviors, particularly in multinational organizations operating across diverse cultural environments.

This project proposes a quantitative methodology to measure corporate culture at Banco Bilbao Vizcaya Argentaria (BBVA) through the construction of a Corporate Culture Index (CCI). Our objective is to transform BBVA's core values, Customer Comes First, We Think Big, and We Are One Team, into measurable indicators and assess how effectively these values are embedded across the organization. The study also examines whether cultural alignment influences key organizational outcomes, including employee retention, engagement, innovation, customer trust, and overall performance across the countries in which BBVA operates.

The framework is built around four main objectives. Firstly, it seeks to translate BBVA's corporate values into observable behavioral indicators. Secondly, it evaluates the degree of alignment between managerial communication and employee perceptions of culture. Thirdly, it investigates differences in cultural implementation across business units. Finally, it analyzes the relationship between cultural alignment and organizational outcomes.

To address important methodological challenges, the proposed model distinguishes between an Internal Culture Index and an External Validation Layer. The Internal Culture Index is constructed exclusively from employee-level and organizational indicators, while the External Validation Layer consists of customer, reputation, and performance outcomes. This separation avoids the circularity that arises when the same variables are used both as components of the index and as measures of its success.

The methodology combines employee survey, HR indicators, organizational metrics, external reputation measures (including Statista Brand KPIs, and LSEG/Refinitiv ESG data), and benchmarking against major Spanish banking peers such as Santander, CaixaBank, Sabadell, Bankinter, and Unicaja. Because access to BBVA's internal data has not been granted, the project adopts a multi-source data generation strategy, including a primary Culture Perception Survey, publicly available external datasets, a synthetic pilot database replicating the expected BBVA data structure, and scenario simulations designed to test whether the index behaves according to theoretical expectations.

To ensure methodological rigor and replicability, the framework incorporates reliability analysis, factor analysis, sensitivity testing, and additional robustness checks. Particular attention is given to challenges such as weighting decisions, indicator overlap, endogeneity, cross-country measurement equivalence, and the risk of conflating corporate culture with employee satisfaction.

The expected outcome is a practical, scalable, and statistically validated measurement system that BBVA can use to monitor cultural implementation, identify perception gaps, benchmark performance, and support strategic decision-making across its international operations.

2. Introduction

Corporate culture is increasingly recognized as a strategic asset capable of influencing organizational performance, employee satisfaction, innovation, and customer trust. In Schein's (2010) classic formulation, organizational culture is the pattern of basic assumptions that a group has developed as it learns to cope with its problems of external adaptation and internal integration. In the banking sector this intangible carries particular weight: supervisory bodies such as the Bank for International Settlements have repeatedly linked weak risk culture to conduct failures and financial instability, while a growing empirical literature connects employee sentiment to firm value.

Traditional approaches measure culture qualitatively, e.g. through interviews, ethnographic observation, or case studies. These methods generate rich insight but are subjective, difficult to compare across units, and provide no standardized metric that management can monitor over time. Organizations therefore require quantitative tools to track culture systematically and to compare outcomes across business units and countries.

BBVA provides an ideal setting for developing such a tool. The bank operates in more than 30 countries, employs roughly 127,000 people (127,174 in FY2025 according to LSEG data, Refinitiv), and serves around 77 million clients. Its international presence creates both the opportunity and the necessity for a scalable framework capable of measuring culture implementation across diverse organizational and national contexts. At the same time, BBVA has articulated a compact and explicit value system, three corporate values with defined expected behaviors, which makes the operationalization problem tractable, therefore, instead of measuring "culture" in the abstract, the project measures the implementation and perception of a specific, stated value system.

It is important to state from the outset what this project is not. A high employee satisfaction score does not automatically imply value alignment; a bank can have content employees and still fail to live up to its stated values. The framework therefore treats satisfaction as one dimension among several, anchors every dimension to a specific BBVA value, and validates the resulting index against external outcomes that were deliberately excluded from its construction.

3. Research Question and Objectives

3.1 Main Research Question and Objectives

How can BBVA's corporate values be translated into measurable indicators, and to what extent does corporate culture influence employee and organizational outcomes across the bank's Spanish regions and business units?

This main question decomposes into the four research questions stated in the project brief:

Objective 1. Develop a Corporate Culture Index that converts BBVA's three corporate values into measurable indicators, with an explicit value-to-proxy mapping that guarantees construct validity.

Objective 2. Measure the degree of alignment between management communication and employee perception of the values, using survey instruments and external review data.

Objective 3. Assess differences in culture implementation within the Spanish market, across regions, departments, and business units, checking that the survey instrument works consistently across groups before scores are compared.

Objective 4. Analyze the relationship between culture and organizational outcomes — retention, innovation, customer trust, and performance, using econometric models that explicitly confront endogeneity and reverse causality.

4. Literature Review

4.1 Defining Corporate Culture

Schein (2010) defines organizational culture as the pattern of shared basic assumptions learned by a group as it solves its problems of external adaptation and internal integration; it is a definition that emphasizes culture as learned, collective, and largely tacit. Hofstede's cultural dimensions theory complements this organizational view with a national one, showing that values such as power distance, uncertainty avoidance, and individualism vary systematically across countries; this is directly relevant for a bank operating in Spain, Mexico, Turkey, Colombia, Argentina, and beyond, because the same survey item may be interpreted differently across national cultures. Guiso, Sapienza and Zingales (2015) provide the economic framing adopted in this project: corporate culture is an intangible asset, and the gap between advertised values and the values employees actually perceive is what determines whether culture creates value. Their finding that perceived integrity, and not proclaimed values, correlates with performance motivates our focus on measuring implementation rather than communication.

4.2 Culture and Employee Outcomes

A substantial literature uses employee sentiment as a measurable proxy for organizational culture. A recent meta-analysis of 137 studies covering 1990–2023 (Hassan et al., 2025) establishes job satisfaction as a core, measurable proxy in banking, with a pooled effect size of $r = 0.41$ (95% CI [0.37, 0.44]) between satisfaction and firm outcomes. Engagement research similarly links psychological safety, leadership trust, and collaboration to retention and discretionary effort. We note, however, the limitation emphasized in our advisor's feedback: satisfaction is an outcome that culture can produce, not culture itself. The literature supports using satisfaction as one indicator within a multidimensional construct, not as a stand-in for the construct.

4.3 Culture and Financial Performance

Evidence connecting culture to financial performance has strengthened as large-scale employee review data have become available. Hassan et al. (2025) report that a one-point rise in satisfaction (on a 5-point scale) is associated with a 0.68–0.73% increase in market capitalization and 0.62–1.01% in annual revenue, controlling for size, leverage, and country. Li and Park (2023), using a panel of 412 listed banks (2012–2022) with firm fixed effects, find that a one-point increase in Glassdoor rating predicts a +0.73% next-quarter stock return ($\beta = 0.0073$, $p < 0.001$), with stronger effects in customer-facing roles; endogeneity is addressed with a two-quarter lag. Guiso, Sapienza and Zingales (2015) reach a consistent conclusion with a different identification strategy. For the banking sector specifically, BIS and OECD reports on governance and risk culture document the supervisory consensus that cultural failures precede conduct and risk failures. Together, this literature justifies treating culture as measurable and economically consequential, while warning that contemporaneous correlations between culture proxies and performance can run in both directions, a concern our econometric strategy addresses with lagged specifications and fixed effects.

5. Conceptual Framework

Corporate culture is built around shared values that guide employee behavior and organizational decision-making. BBVA defines its culture through three core values. To measure culture quantitatively, these values must be translated into observable behaviors and measurable indicators.

5.1 BBVA Values and Expected Behaviors

Customer Comes First. Expected behaviors: employees prioritize customer needs in decision-making, resolve problems end-to-end rather than passing them on, and treat customer trust as the primary asset of the bank. Observable manifestations: customer satisfaction and trust scores, customer NPS, attrition, complaint resolution times, and the customer-orientation items of the employee survey.

We Think Big. Expected behaviors: employees experiment, propose improvements, pursue continuous learning, and embrace the bank's digital transformation agenda. Observable manifestations: ideas submitted per 100 employees, training completion and training hours, internal mobility, and promotion rates.

We Are One Team. Expected behaviors: employees collaborate across functions and geographies, trust their leaders, and feel safe to raise concerns. Observable manifestations: collaboration scores, leadership trust (360 degree feedback), psychological safety scales, and engagement survey participation.

5.2 Value-to-Proxy Mapping

BBVA Value	Culture Dimension(s)	Possible Measurable Proxies
Customer Comes First	Customers (external validation layer)	Customer NPS - Brand trust (Statista) · Customer attrition - Customer satisfaction
We Think Big	Innovation - Employee Experience	Ideas per 100 FTE - Training completion & hours - Internal mobility - Promotion rates
We Are One Team	Engagement - Satisfaction - Ethics	Collaboration scores - Leadership 360° trust - Psychological safety - eNPS - Compliance incidents

Table 1. Mapping of BBVA's three corporate values to culture dimensions and measurable proxies.

5.3 Conceptual Model

The model assumes a causal chain running from stated values to business performance:

BBVA Values → Employee Behaviors → Culture Dimensions → Employee Outcomes → Customer Outcomes → Business Performance

Two design decisions follow from this chain. First, variables located at different points of the chain must not be mixed inside a single index: customer outcomes are manifestations or consequences of culture, not components of it. They are therefore moved to an External Validation Layer (Section 7). Second, the chain clarifies the distinction between culture and satisfaction: satisfaction sits among employee outcomes, so it can serve as one indicator of how culture is experienced, but the index also includes behavioral indicators (participation, absenteeism, turnover, idea submission) that are harder to game and conceptually closer to culture itself.

6. BBVA Corporate Culture and Values

6.1 Current Value System

BBVA's corporate culture's values are supported by the bank's broader identity as a digital transformation leader in banking and its commitment to sustainable finance. The values are communicated through leadership messaging, internal channels, onboarding, and the bank's public reporting.

6.2 Current Measurement Practices

Like most large banks, BBVA currently monitors culture-adjacent indicators through annual employee engagement surveys, quarterly eNPS pulse surveys, leadership 360 degree feedback, compliance and conduct incident logs, and HR records on turnover, absenteeism, mobility, and training. Externally, culture-adjacent signals also come from Statista brand KPIs, and ESG data providers such as LSEG/Refinitiv. What is missing, and what this project supplies, is a methodology that integrates these scattered signals into a single, validated, comparable index linked explicitly to the three values.

The LSEG ESG data already collected for this project (Annex B) gives a first observed picture: BBVA reports an employee satisfaction score of 90.0% in FY2025 (89.2% in FY2024), employee turnover of 11.8%, 57.5 average training hours per employee with a training investment of approximately €495–582 per employee, 51.4% women employees and 46.9% women managers, and a Social Pillar ESG score of A. These figures are favorable but illustrate exactly the measurement problem this project addresses: they are disconnected single indicators, not a coherent measure of value implementation.

6.3 Identified Perception Gaps

External (customers). Statista Brand KPI data (2024, US sample, n = 1,200, 0–10 scale) indicate that BBVA outperforms peers on Cleverness (7.8 vs. 6.9, +0.9) and Value (7.5 vs. 6.8, +0.7) but underperforms on attributes that customers rate as most important, including Trust (5.9 vs. 7.2, −1.3), Reliability (6.1 vs. 7.0, −0.9), and Friendliness (6.4 vs. 7.1, −0.7). The peer average is calculated as the mean of JPMorgan, Citi, Bank of America, and Santander US. These findings suggest that BBVA is perceived as innovative and capable but has not yet achieved the same level of trust and reliability as key competitors, creating a potential tension with the value *Customer Comes First*.

Internal (employees). Internal employee engagement and culture data were not made available for this study. As a result, no direct assessment of employee perceptions could be conducted. Any discussion of internal culture gaps is therefore limited to publicly available sources, including employee reviews and external workplace research, which indicate generally positive engagement and brand pride but also recurring concerns regarding leadership trust, incentive structures, and psychological safety. These observations should be treated as indicative rather than conclusive and cannot be considered representative of the broader employee population. Consequently, the analysis focuses primarily on externally observable perception gaps, while considering the limitations created by the absence of internal survey data.

7. Methodology

This section sets out the measurement approach and the statistical models, and explains how we arrived at them. The methodology answers a deceptively simple question (how do you put a number on something as intangible as culture?) in four moves: define what the number is built from (the index and its dimensions), decide how the ingredients are combined (standardization and weights), show the arithmetic on a worked example, and specify the models that turn the resulting index into evidence about outcomes.

7.1 How We Arrived at This Approach

Our design was shaped by three constraints. The first constraint is conceptual: culture is not the same as satisfaction, so a credible measure cannot be a single happiness score, it must be multidimensional and tied explicitly to BBVA's stated values (Section 5). The second is methodological: an early single-index draft mixed cultural inputs with their own consequences (customer outcomes, turnover), which makes any later validation circular; the fix was to split the framework into an internal index and a separate external validation layer (Section 7.2). The third is practical: with no access to internal data, the index has to be buildable from a survey the team can run plus public benchmarks, which is what drives the variable choices in Section 7.3. Additionally, after some consideration, we added a fourth principle: keep weighting choices transparent and testable rather than asserted, which is why the weights are presented as pilot assumptions and stress-tested rather than fixed (Sections 7.6 and 11, now folded into the advised steps of Section 11). The result is a methodology designed to be adjustable: every choice that could reasonably be made differently is exposed as a parameter the user can change.

7.2 Index Construction Logic

Our initial plan proposed a single index with six dimensions: Ethics (20%), Satisfaction (20%), Engagement (20%), Employees (15%), Customers (15%), and Innovation (10%). The internal review identified two structural problems with this design. First, circularity: variables such as turnover, customer attrition, and customer NPS appeared both as inputs to the index and as outcome variables in the validation regressions, which weakens any causal interpretation. Second, mixing of causes and outcomes: combining internal cultural indicators with external customer outcomes inside one index makes it impossible to determine whether customer perceptions are drivers, manifestations, or consequences of culture.

The revised architecture therefore splits the framework into two layers:

(a) Internal Culture Index (ICI). Built exclusively from employee and organizational indicators across five dimensions: Ethics, Satisfaction, Engagement, Employee Experience, and Innovation. The original six-dimension weights are rebased proportionally onto the five internal dimensions, and presented explicitly **as pilot assumptions, not final estimates.**

(b) External Validation Layer. Customer and reputation outcomes, such as brand trust, customer NPS, customer attrition, and external reputation measures, have now been removed from the index and used only to validate it. A valid culture index should predict these outcomes; it must not contain them.

Dimension	Pilot weight (6-dim draft)	Rebased internal weight (ICI)	Linked BBVA value
Ethics	20%	23.5%	We Are One Team / integrity
Satisfaction	20%	23.5%	We Are One Team
Engagement	20%	23.5%	We Are One Team
Employee Experience	15%	17.6%	We Think Big
Innovation	10%	11.8%	We Think Big
Customers	15%	— (moved to validation layer)	Customer Comes First

Table 2. Pilot six-dimension weights rebased onto the five internal dimensions after moving Customers to the validation layer.

For now, there is no strong theoretical rationale for why Ethics should weigh twice as much as Innovation; the weighting scheme is therefore treated **as a pilot specification,** and Section 7.6 and Section 11 describe how alternative weighting schemes will be estimated and compared.

7.3 Variable Selection

Each observation is one employee-year (or bank-year for the benchmark module). By using no-internal-data strategy, every variable is sourced from one of three channels: the team's own Culture Perception Survey (Section 8.2), the synthetic three-scenario generation (Section 8.4), or observed public data, such as the Refinitiv Public Companies dataset, Statista, and annual and ESG reports (Section 8.3).

The pilot variable set is:

Variable	Source (no-internal-data strategy)	Scale	Direction	Dimension
Compliance incidents per 100 FTE	Synthetic (pilot); bank-level proxy: conduct fines and sanctions from annual/ESG reports	Count	−	Ethics
Ethics climate score	Survey item (Culture Perception Survey)	0–10	+	Ethics
eNPS	Survey item ("How likely are you to recommend...?", promoters vs detractors)	−100 to +100	+	Satisfaction
Perceived pay equity	Survey item; bank-level anchor: Refinitiv gender pay-gap ratio	0–1	+	Satisfaction
Survey participation rate	Observed by the team (response rate of own survey, by department)	%	+	Engagement
Absenteeism rate	Synthetic (pilot only — HR-system variable, not obtainable)	%	−	Engagement

Psychological safety scale	Survey items (adapted Edmondson scale)	1–5	+	Engagement
Turnover rate	Synthetic at employee/department level; observed at bank level (Refinitiv: BBVA 11.8%)	%	−	Employee Experience
Internal mobility	Survey self-report ("changed role internally in the last 2 years"); synthetic in pilot	%	+	Employee Experience
Leadership trust score	Survey items (replaces the original 360° system measure)	1–5	+	Employee Experience
Ideas submitted per 100 FTE	Synthetic (pilot); survey self-report alternative ("submitted an improvement idea this year")	Count	+	Innovation
Training completion / hours	Survey self-report; bank-level anchor: Refinitiv average training hours	%	+	Innovation
Brand trust score	Statista Brand KPIs (study 190428), observed	0–10	+	Validation layer
Customer NPS	Public reports / synthetic in pilot	−100 to +100	+	Validation layer
Customer attrition rate	Annual reports, observed where disclosed	%	−	Validation layer
Region, department, gender, tenure band, compensation band	Survey demographics (categorical bands, anonymous)	Categoric al	—	Controls

Table 3. Pilot variable set: each indicator, its data source under the no-internal-data strategy, scale, direction, and culture dimension. Direction: + = higher value indicates stronger culture; − = reverse-coded before entering the index.

The index splits into a perception block, fully measurable today through the survey (ethics climate, eNPS, pay equity, psychological safety, leadership trust, mobility, training, and idea-submission self-reports), and a behavioral block (absenteeism, department turnover, compliance incidents), which exists only synthetically until an organization applies the framework on its own records. A leave-one-block-out comparison is therefore added to the robustness battery to show how much of the index survives on survey data alone. And before constructing an index, the candidate indicators are advised to be screened for overlap: satisfaction, engagement, leadership trust, and psychological safety are likely to be highly correlated, so a correlation matrix and factor analysis would help identify redundant indicators, which are dropped or merged rather than double-counted.

7.4 Standardization

To ensure comparability across indicators measured in different units, all variables are standardized to a common scale ranging from 0 to 1 using min–max normalization. The standardized score is calculated as: $X = (X - \min) / (\max - \min)$, where *min* and *max* represent the lower and upper bounds of the relevant peer-group benchmark range. For indicators where lower values indicate better performance, such as turnover, absenteeism, compliance incidents, and customer attrition, reverse normalization is applied: $X = 1 - [(X - \min) / (\max - \min)]$

This transformation ensures that higher values consistently reflect stronger organizational culture across all dimensions. For example, an eNPS score of +12 within a benchmark range of −30 to +45 yields a standardized score of 0.56. Similarly, a turnover rate of 18% within a benchmark range of 8% to 30% produces a reverse-coded score of 0.55. Standardizing all indicators in this manner allows them to be directly compared and aggregated into a composite culture index.

7.5 Aggregation

$$\textbf{Culture Index} = \Sigma\,(w_i \times X_i^*) \times 100$$

Dimension scores are the weighted means of their standardized indicators; the index is the weighted sum of dimension scores, rescaled to 0–100 (divide by 100 for the normalized 0–1 version). A full step-by-step computation, including the corrected version of the pilot deck's illustrative example and a partial computation on observed LSEG benchmark data, is provided in Section 7.7. These numbers are demonstration values, not estimates of BBVA's actual culture.

7.6 Alternative Weighting Schemes

Because the pilot weights are theory-based rather than empirically estimated, the final report will compare results across four specifications: (i) equal weights; (ii) PCA-derived weights, where the first principal component's loadings determine relative importance; (iii) CFA-derived weights, estimated in R once real data arrive; and (iv) expert-weighted versions elicited from BBVA Talent & Culture. A robust index should produce broadly consistent rankings across all four; substantial divergence would itself be an important finding about the fragility of composite culture measures.

7.7 Worked Calculations

7.7.1 Worked Example A. Employee-Level Illustration (Demonstration Values)

The following computation reproduces, step by step, the illustrative example from the pilot deck, using demonstration values for a hypothetical BBVA employee-year observation. Step 1, standardize: eNPS of +12 against a sector range of [−30, +45] gives $eNPS^* = (12 - (-30)) / (45 - (-30)) = 42/75 = 0.56$, i.e., the 56th percentile of the peer group. Step 2, apply reverse-code: a turnover rate of 18% against a sector range of [8%, 30%] gives $Turnover^* = 1 - (18 - 8)/(30 - 8) = 1 - 0.45 = 0.55$. Step 3, aggregate:

Dimension	Standardized score X*	Weight (6-dim pilot)	Contribution w × X*	Weight (revised ICI)	Contribution w × X*
Ethics	0.68	0.20	0.1360	0.235	0.1598
Satisfaction	0.56	0.20	0.1120	0.235	0.1316
Engagement	0.61	0.20	0.1220	0.235	0.1434
Employee Experience	0.55	0.15	0.0825	0.176	0.0968
Customers	0.47	0.15	0.0705	— (validation layer)	—
Innovation	0.52	0.10	0.0520	0.118	0.0614
Index (Σ × 100)		1.00	57.5	1.00	59.3

Table 4. Worked Example A. Employee-level index computation under the pilot and rebased weights (demonstration values).

Two observations follow. First, the correct weighted sum of the deck's demonstration values is 0.575, i.e., an index of 57.5; the figure of 58.4 reported in the pilot presentation contained a small arithmetic slip, corrected here. Second, removing the customer dimension and rebasing the weights changes the score from 57.5 to 59.3 = a 1.8-point shift in this example. The leave-one-dimension-out analysis (Section 11) generalizes exactly this comparison.

7.7.2 Worked Example B. Bank-Level Computation with Observed Refinitiv LSEG Data

A partial, bank-level version of the index can already be computed with the observed FY2025 data collected from LSEG Workspace (Annex B). Three of the five internal dimensions have observable proxies in that extraction: Satisfaction (reported employee satisfaction), Employee Experience (employee turnover), and Innovation/development (average training hours). Engagement and Ethics have no comparable peer-level proxy in the extraction and are omitted; the remaining weights are rebased to sum to one (23.5/52.9 = 0.444, 17.6/52.9 = 0.333, 11.8/52.9 = 0.223).

Indicator	BBVA raw value	Benchmark min	Benchmark max	BBVA standardized X*	Rebased weight	Contribution
Employee satisfaction (%)	90.0	85.0 (Santander)	90.0 (BBVA)	(90−85)/(90−85) = 1.000	0.444	0.444
Employee turnover (%) — reverse	11.8	8.0 (sector assumption)	30.0 (sector assumption)	1 − (11.8−8)/22 = 0.827	0.333	0.275
Avg. training hours / employee	57.5	22.5 (Sabadell)	65.9 (CaixaBank)	(57.5−22.5)/43.4 = 0.806	0.223	0.180
Partial observed ICI					1.000	0.899 or 89.9

Table 5. Worked Example B. Partial bank-level index for BBVA computed from observed Refinitiv data (three observable dimensions).

BBVA's partial observed index of 89.9 should be read as a methodological demonstration, and its very magnitude is instructive. The score is inflated for three identifiable reasons: (i) only three of five dimensions are observable, and the missing ones (Ethics, Engagement) are precisely those where internal perception gaps are hypothesized; (ii) BBVA sits at the maximum of the satisfaction indicator within a peer set of only three reporting banks, so min–max scaling assigns it a perfect 1.000 by construction, which is the boundary-instability problem of small benchmark samples discussed in Section 10; and (iii) the turnover range is a sector assumption rather than an observed peer range, because peer turnover is not disclosed in the extraction. Applying the same procedure to Banco Sabadell (satisfaction 86.6 → 0.32; training hours 22.5 → 0.00; turnover unavailable, weights rebased over the two remaining indicators) yields a partial score of 21.3 = a 68-point gap between two banks whose underlying raw differences are modest. This contrast quantifies why the final design (a) expands the benchmark set, (b) reports confidence intervals around benchmark rankings, and (c) compares min–max scaling against z-score and percentile alternatives before any ranking is published.

7.8 Econometric Models

Once the Culture Index is computed, the next question is whether it is actually connected to outcomes that matter: do people stay, are they satisfied, do they propose ideas, do banks with stronger culture do better? All five models answer a version of that question, and all follow one simple template:

$$Outcome = a + b \times Culture\ Index + control\ variables$$

The number b says how much the outcome changes when the Culture Index is one point higher, holding everything else fixed. The control variables (gender, tenure, pay band, region, department, year, regional unemployment) strip out differences that have nothing to do with culture, for example, that exits are higher in regions with a hotter job market. Three precautions apply to every model: using the previous year's index, so culture is measured before the outcome it is supposed to explain; use\ing statistical methods that do not overstate our certainty when observations within a department resemble each other; and all estimation runs on the analysis universe (the Average scenario plus the peer benchmark), never on the Strong/Weak replications, which are reserved for validation.

Model 1: Does stronger culture keep people from leaving?

$$\text{Probability(employee leaves)} = \Lambda(\ a + b \times Culture\ Index_{last\ year} + controls\ )$$

Reading the formula: Λ (the logistic function) is simply the device that converts the right-hand side into a probability between 0% and 100%, i.e. the adaptation needed because the outcome here is a yes/no event (whether an employee left during the year), known as logistic regression. B is expected to be negative: a higher culture score should mean a lower chance of leaving. The result also has a direct money interpretation: if a 10-point higher index goes together with a 2-percentage-point lower exit probability, then at BBVA Spain's scale (≈23,000 employees, ≈€18,000 of hiring and lost-productivity cost per avoided exit) that difference is worth roughly €8 million per year.

Model 2: Does culture show up in satisfaction?

$$eNPS = a_{employee} + b \times Index\ excl.\ Satisfaction_{last\ year} + controls + error$$

Reading the formula: each employee gets their own personal baseline (the a with the employee subscript), which absorbs stable personality differences, some people simply rate everything higher , so b is identified from changes, not from comparing optimists to pessimists. The outcome is the

employee's eNPS. There is an obvious trap here: satisfaction is itself one of the five dimensions inside the index, so the index would partly be predicting itself. To avoid this circularity, the index on the right-hand side is recomputed without the Satisfaction dimension, so the model asks whether the other four dimensions (ethics, engagement, employee experience, innovation) predict how satisfied people say they are. B is expected to be positive.

Model 3: Does culture spark ideas?

$$\text{Expected ideas} = (FTE / 100) \times \exp(a + b \times Culture\ Index_{last\ year} + controls)$$

Reading the formula: the exponential keeps predictions positive (a count of ideas can never be below zero), and the FTE/100 term simply adjusts for department size, so b is interpreted per 100 employees, specifically, exp(b) is the multiplier on the idea rate for each extra index point. The outcome is the number of ideas submitted per 100 employees, measured at department level. Counts cannot be negative and tend to pile up in a few very active departments, so the template is adapted to count data (a Poisson model, switched to its more flexible cousin, the Negative Binomial, if the idea counts are more spread out than the simple model assumes). B is expected to be positive: departments with stronger culture should generate more ideas.

Model 4: Do banks with stronger culture perform better?

$$Bank\ result = a_{bank} + b \times Bank\ Culture\ Index_{last\ year} + controls + error$$

Reading the formula: each bank gets its own baseline (the a with the bank subscript), so b compares each bank with its own past rather than mixing up permanent differences between, say, Santander and Sabadell. The outcome is bank-level performance, which is return on assets, revenue growth, brand trust, on the five-bank Spanish panel, using the observed Refinitiv financials. Here honesty requires a warning: with only five banks, no statistical test can be conclusive. The model is reported as a co-movement check– do banks that score higher on the index also tend to show better outcomes? — and never as proof.

Model 5 : Do Spanish regions differ? (In case the bank has regional data)

$$Culture\ score = overall\ mean + region\ effect + department\ effect + individual\ part$$

$$ICC_{region} = \text{Var}(region) / [\ \text{Var}(region) + \text{Var}(department) + \text{Var}(individual)\ ]$$

Reading the formulas: every employee's culture score is decomposed into the company-wide average plus a piece shared by everyone in the same region, a piece shared by everyone in the same department, and a purely individual piece. The ICC is then just a ratio: of all the variation in scores, what fraction is the regional piece? This model answers Research Question 3 and works differently: instead of predicting an outcome, it splits the total variation in employees' culture scores into three layers — how much is between regions, how much between departments, and how much between individuals. The share sitting at the region level (called the intraclass correlation, ICC) is the answer: if regions account for a meaningful share, a common rule of thumb is more than 10%, then culture implementation differs systematically across Spain and regional action plans are justified; if the share is small, culture is mostly a department-and-individual story. In the pilot data the regional share for BBVA is about 7%, i.e., visible but modest, which is what one would expect within a single country.

Summary of Hypotheses

What we expect	Model	What would confirm it
H1: Where culture is stronger, fewer people leave	Model 1	b negative and statistically significant
H2: Where culture is stronger, satisfaction (eNPS) is higher	Model 2	b positive and statistically significant
H3: Where culture is stronger, more ideas are submitted	Model 3	b positive and statistically significant
H4: Banks with stronger culture show better results	Model 4	positive co-movement (read cautiously: only 5 banks)
H5: Culture implementation differs across Spanish regions	Model 5	region share of variation above ≈10%

Table 6. The five hypotheses, the model that tests each, and the result that would confirm it.

7.9 Expected Relationships

Based on our methodology and subject to validation on real data, we expect stronger culture to be associated with lower voluntary turnover, higher engagement and eNPS, higher customer trust, and better organizational performance. Each hypothesis is linked to an index dimension and a measurable KPI, producing six strategic levers for BBVA:

Area	Possible Problem	What BBVA Should Do	Expected Improvement
Ethics	Managers don't act according to company values	Include ethics in manager evaluations and bonuses	Fewer compliance issues and better ethics ratings
Employee Satisfaction	Employees feel pressure to chase revenue instead of doing good work	Tie part of bonuses to customer satisfaction and risk management	Employees become more satisfied (higher eNPS)
Engagement	Employees don't participate and absenteeism is high	Run monthly culture surveys and hold managers accountable	More participation, less absenteeism
Employee Experience	Employees feel they have no voice and leave the company	Give teams more decision-making power and hold regular conversations with senior leaders	Lower employee turnover
Customer Perception	Internal culture problems affect customers' trust	Improve customer journeys and track customer experience metrics	Higher customer trust and better NPS
Innovation	Employees don't feel safe sharing ideas	Create anonymous idea channels and invest more in training	More ideas generated and higher training participation

Table 7. Six strategic levers for BBVA

The €4.2M figure is derived as follows: BBVA Spain headcount ≈ 23,000; industry-average cost-to-hire plus productivity loss per exit ≈ 60% of annual salary (≈€18k on a €30k average); one percentage point of turnover avoided ≈ 230 exits × €18k ≈ €4.1–4.2M. All targets are illustrative and will be re-estimated on real data.

One caution applies to everything above: correlation is not causation. Banks that perform well may also make employees happier (reverse causality), and pay or local job markets may move both culture scores and outcomes at the same time. Using last year's index, adding controls, and comparing within regions and departments reduces these problems but does not eliminate them, so the final report will describe associations, not causal effects, unless much stronger evidence becomes available

8. Data Description and Strategy

8.1 Overview

The project has no access to BBVA internal data, HR system records, engagement survey results, compliance logs, or turnover files. Thus, they are not assumed at any stage of the analysis. The data strategy is therefore built entirely on sources that the team can actually obtain, i.e. no-internal-data strategy: survey data generated directly by the team, external public data, such as Refinitiv dataset, and artificially generated data. The framework is nevertheless designed so that, if BBVA chooses to apply it internally in the future, they could use their own records to replace the survey and synthetic components without changing the methodology, but no part of the present project relies on that happening.

8.2 Survey Data

The only employee-level source is Culture Perception Survey, distributed via Microsoft Forms to artificial BBVA Spain employees. The survey generates the perception-based indicators of the index: trust in leadership, psychological safety, value alignment, engagement, team collaboration, customer orientation, innovation culture, ethics climate, and satisfaction and well-being (eNPS item included). Respondent attributes, region, department, gender, tenure band, are collected as categorical bands so that no individual is identifiable, and responses are anonymous. The target is a minimum viable sample of 500 responses; until the survey is fielded and reaches sufficient scale, its structure is emulated by the artificial data of Section 8.4.

8.3 External Public Data

External data populate the bank-level benchmark and the external validation layer. The principal source is the Refinitiv Public Companies dataset (accessed via LSEG Workspace), from which a first benchmark extraction covering BBVA and the Spanish peer banks (Santander, CaixaBank, Sabadell, Bankinter, Unicaja) has already been assembled: employee satisfaction, turnover, training hours and cost, workforce diversity, ESG pillar scores, and full financial screening (ROA, market capitalization, P/E, P/B, dividend yield). This is complemented by Statista Brand KPIs (study 190428) for customer-side perception (trust, reliability, friendliness, value, cleverness), Glassdoor reviews and ratings, Great Place to Work rankings, and the banks' published annual and ESG reports for compliance fines and innovation disclosures. These sources are genuinely observed, bank-level, and publicly verifiable; they anchor the min–max benchmark ranges and provide the outcomes for Model 4.

8.4 Artificial Data Generation: Three-Scenario Replication

Under the no-internal-data strategy, the employee-level dataset is artificially generated and replicated under three possible scenarios. A panel of 500 simulated BBVA Spain employees is drawn once, fixing identifiers, region, department, gender, tenure, compensation band, and individual effects, and then replicated three times, with every indicator regenerated under a different assumed culture regime: Scenario A (Strong culture: high trust and engagement, low turnover, high innovation), Scenario B (Average culture: moderate scores, calibrated to the observed public anchors from the Refinitiv dataset, such as BBVA's reported 11.8% turnover and 90% satisfaction), and Scenario C (Weak culture: low trust and engagement, high turnover, elevated compliance incidents). This implements the AI-assisted survey-replication approach from the team's work plan: the same simulated respondent pool is "re-surveyed" under three counterfactual organizational conditions. The Average scenario serves as the baseline for estimating Models 1–5; the Strong and Weak replications verify that the index and the models respond as theoretically expected when the underlying regime changes. Once real survey responses arrive, they progressively replace the synthetic baseline while the scenario replications are retained as a stress test of the methodology.

8.5 Indicator Sourcing Without Internal Data

Each of the twelve index indicators is mapped to a source the team actually has. Perception indicators (ethics score, eNPS, leadership trust, psychological safety, pay-equity perception, training completion) come from survey items, emulated synthetically in the pilot. Indicators that would normally come from HR systems (compliance incidents, absenteeism, department turnover, internal mobility, participation rates, ideas submitted) exist only synthetically at the employee and department level; at the bank level they are proxied where possible by Refinitiv disclosures (turnover, training) and annual-report figures.

To better understand how much the missing data for HR affects our results, we decided to test what happens to the index when the behavioral block is removed entirely from the analysis. When only perception indicators are kept there is a significant average scenario index drop from 47,9 to 43.2 a difference of around 4,7 points. This suggests that even without internal access to HR records, the survey based perception block solely captures most of the index's information, which proves the right choice of following the no-internal-data strategy. The behavioral indicators still matter though, especially for Ethics and Engagement, where survey responses and actual HR data can tell very different stories. We are transparent about this limitation: because this part of the index is built on synthetic data, the results are illustrative rather than conclusive. Real findings about BBVA would only be possible with a larger survey rollout or access to internal data we did not have.

8.6 The Synthetic Employee Data and How It Behaves

Figure 1 summarizes the artificially generated employee data: it shows the aggregate value of each indicator under the three culture scenarios and the peer benchmark. Reading it is the quickest way to see both what the synthetic data contain and why the generation design is trustworthy.

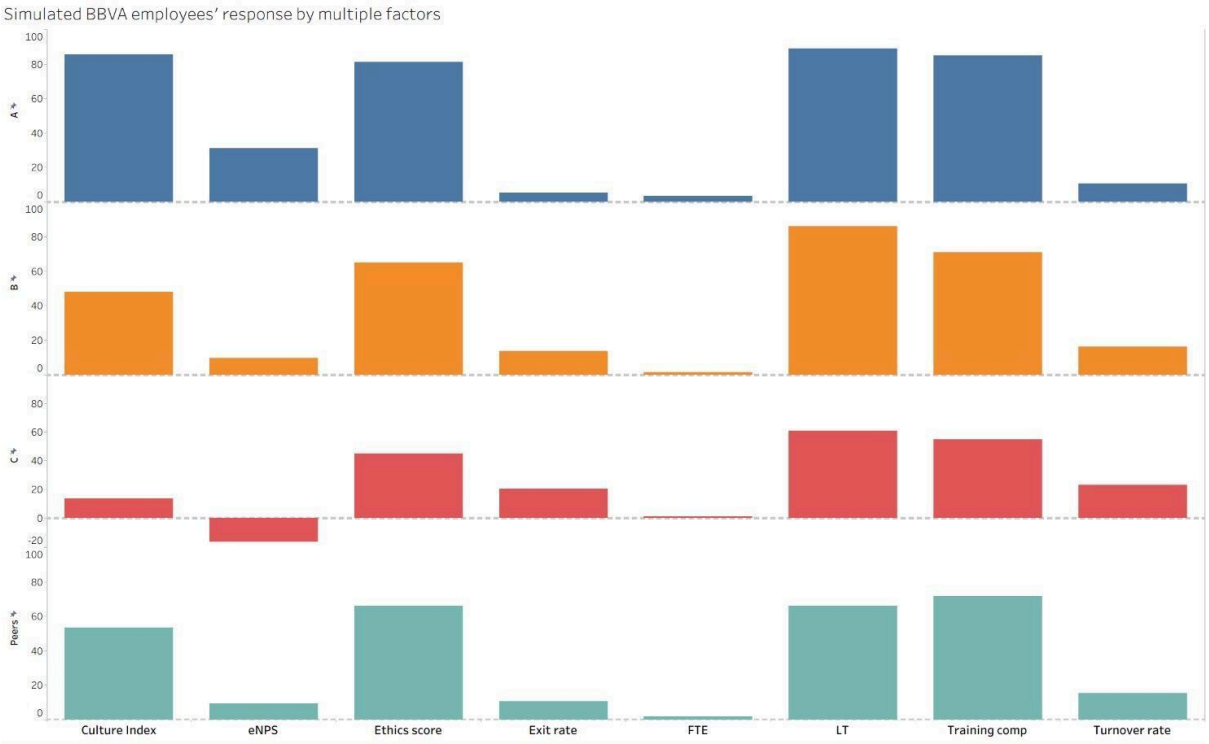


Figure 1. Simulated BBVA employees' responses by multiple factors: aggregate means of the three scenario replications (A-Strong, B-Average, C-Weak) and the Spanish peer benchmark.

Each row of the chart is one group of the artificially generated employee dataset of Section 8.4: Scenario A (Strong culture, blue), Scenario B (Average culture, orange), Scenario C (Weak culture, red), and the pooled peer-bank Benchmark (green). The same 500 simulated BBVA employees underlie rows A, B, and C, only the assumed culture regime changes between them. Each column is the aggregate mean of one variable from the employee-year file: the Culture Index (0–100), eNPS (−100 to +100), the ethics climate score, the exit rate, FTE (a composition check, see below), leadership trust (LT), training completion, and the turnover rate. The ethics and leadership-trust scores are rescaled to share the common 0–100 axis with the other metrics.

Every metric where higher means stronger culture falls steadily from row A to row C: the Culture Index drops from about 85 to 48 to 13, eNPS from roughly +31 to +10 to −17, turning negative in the weak scenario, meaning detractors outnumber promoters, and ethics, leadership trust, and training completion follow the same staircase. The two reversed metrics move the opposite way, exactly as they should: the exit rate climbs from about 5% to 14% to 20% and turnover from about 9% to 16% to 23% as culture weakens. The Benchmark row sits close to Scenario B on most metrics, by design: the Average scenario is calibrated so that simulated BBVA lands mid-range among its Spanish peers, consistent with the bank-level results of Section 9.2.

This figure is the visual counterpart of the scenario-validation check in Section 11: it demonstrates, in one view, that the index and every indicator feeding it respond in the right direction and by sensible magnitudes when the underlying culture regime changes, the monotonic A > B > C ordering that the conceptual model of Section 5 requires. The gaps between rows are large relative to the within-row noise, which is what allows the statistical models of Section 10 to detect culture effects in the pilot.

As clarified in section 8.5, the bars summarize simulated data and say nothing about real BBVA employees. The FTE column is flat near zero across all rows because the replications are identical in size and composition by construction; it carries no cultural information and serves only as a sanity check that differences between rows come from the regime, not from the sample. Finally, it is also relevant that the bars show means only: they conceal the variation across regions and departments within each scenario, which is exactly what Model 5 analyzes; a box-plot version of this figure is the natural companion when that within-scenario spread matters.

8.7 Proposed Dataset Structure

8.7.1 Hierarchical Design

The project combines employee-level, department-level, and bank-level variables within the Spanish market. The originally proposed country level is replaced by a generated within-Spain regional level (Madrid, Catalunya, Andalucía, País Vasco, and Comunidad Valenciana), while keeping the scope national. The variables to analyze are thus: Employee ID, Country of Origin, Department, Employee Satisfaction, Employee Engagement, and Employee Retention.

8.7.2 Employee-Year Dataset (Artificially Generated and Replicated)

One row corresponds to one employee-year. The dataset contains no real employee records: it consists of the three-scenario artificial replication described in Section 8.4, which are the same 500 simulated BBVA Spain employees observed in 2024 and 2025 under the Strong, Average, and Weak regimes, Additionally, a single Benchmark realization of 100 simulated employees in each of the four Spanish peer banks, for a master file of roughly 3,800 rows ($500 \times 2 \times 3 + 4 \times 100 \times 2$) and 31 columns, is added: 12 culture indicators, 6 controls, 4 identifiers, 6 derived columns, the scenario tag, an analysis-universe flag, and an aggregation key. Identifiers are SHA-256 hashes, identical across the three replications of the same simulated employee.

The Average scenario plus the Benchmark constitutes the analysis universe: scaling bounds are computed over it, and Models 1–5 are estimated on it; the Strong and Weak replications, scored against the same bounds, deliver the scenario validation (aggregate index 85.4 / 47.9 / 13.4; exit rates 5.2% / 13.7% / 20.4% — the monotonic ordering the conceptual model requires). When the Culture Perception Survey is fielded, its responses adopt this same row structure and replace the synthetic Average panel.

8.7.3 Department-Year and Bank-Year Datasets

The department-year dataset (one row per bank-department-year of the analysis universe, 50 rows) holds the rate variables turnover, absenteeism, participation, mobility, compliance incidents per 100 FTE, and ideas per 100 FTE, and serves as the unit of analysis for Model 3. Because these rates originate in HR systems the team cannot access, this dataset is entirely simulated and exists to validate the pipeline. The bank-year dataset (10 rows) is the opposite case: it is built almost entirely from observed public data: the aggregated Culture Index (BBVA from the Average replication), the external validation layer (Statista brand trust, customer NPS, attrition, Glassdoor rating, excluded by design from index construction), and FY2025 financials (ROA, market capitalization, P/E, P/B) taken directly from the Refinitiv Public Companies dataset for all five Spanish banks.

8.7.4 Pilot Implementation of the Dataset

The structure is implemented in the project file BBVA_Dataset.xlsx (documented sheet by sheet in Section 9). Derived columns are live formulas, so benchmark bounds, reverse coding, and the weighted aggregation under the pilot weights (23.5/23.5/23.5/17.6/11.8) recalculate when any input or weight changes, therefore, the weight-sensitivity checks of Section 11 are executable directly in the workbook (the file itself is described in Section 9). Generation is reproducible (NumPy, seed 42), a codebook documents every variable including its source under the no-internal-data strategy of Section 8.5, a scenario-summary sheet reports the live three-scenario comparison, and the only observed data in the file are isolated on the Refinitiv benchmark sheet and in source-flagged cells of the bank-year panel.

8.8 Observed Data: the Refinitiv Extraction (real_data.xlsx)

This workbook is the raw, unprocessed extraction from the Refinitiv Public Companies dataset (LSEG Workspace, retrieved 2026) and constitutes the project's observed public-data collection (Section 8.3) in its original form. It is the only fully observed data source of the project: every figure in it refers to real, publicly reported company information. The cleaned subset used in the analysis (Table 7 below and the lseg_observed sheet of the workbook in Section 9) is derived from this file; it is preserved unmodified so that every cleaned number can be traced back to its source.

Sheet	Contents	Role in the project
Diversity and Inclusion	LSEG D&I scoring for the Spanish banks: overall D&I score and its four pillars (Diversity, Inclusion, People Development, Controversies), plus workforce indicators (women employees, women managers, board diversity)	Peer benchmark for the social indicators of Annex B; calibration anchors
Other banks vs bbva	Peer screen of Spanish-listed banks: price, market capitalization, P/E, P/B, ROA (FY1), dividend yield for BBVA, Santander, Sabadell, Bankinter, CaixaBank, Unicaja and others	Observed financials of the bank_year panel; outcomes for Model 4
ESG and peer analysis in images	Chart exports from LSEG Workspace (ESG score comparisons across peers)	Visual reference only
Environment	LSEG ESG Statement, Environment pillar, BBVA, FY2021–FY2025 (emissions, resource use, environmental targets)	Context; not used in the Culture Index
Social	LSEG ESG Statement, Social pillar, BBVA, multi-year: employees, employee satisfaction, turnover, training hours and cost, gender pay gap, lost days, disabilities, Social Pillar score	Primary source of the BBVA time series in Annex B and of the calibration anchors (turnover 11.8%, satisfaction 90%, training 57.5 h)
Governance	LSEG ESG Statement, Governance pillar, BBVA, multi-year (board structure, compensation policy, shareholder rights)	Context; supports the Ethics dimension discussion
Controversies 1 / Controversies 2	LSEG ESG Controversies statement and its underlying score detail (ESG score A / 89.0 in FY2025; A+ / 92.0 in FY2024)	Public proxy for the conduct side of the Ethics dimension (Section 7.2)
Financial Summary	Company fundamentals for BBVA (banking-services template): income statement, balance sheet, and key ratios	Background financials; cross-check for Model 4 inputs

Table 8. Sheets of the raw Refinitiv extraction (real_data.xlsx) and their role in the project.

Citation: Refinitiv (LSEG) (2026), Public Companies dataset, ESG statements, social and diversity indicators, and peer financial screening, accessed via LSEG Workspace (full entry in Section 15).

9. Implementation and Results

This section shows the methodology of Section 7 put into practice: the workbook that implements the index and the models, how to use and adjust it, and the results the pilot produces, followed by an honest reading of what those results do and do not establish. Two presentation figures (Figures 2 and 3) visualize the bank-level results, and the closing subsection interprets every number against the project's no-internal-data design.

9.1 The Pilot Analysis Workbook (BBVA_Dataset.xlsx)

This workbook is the working implementation of the dataset design of Section 8 and the methodology of Section 7: the artificially generated, three-scenario replicated employee panel for the Spanish market, together with the live index machinery, the aggregation layers for the models of Section 7.8, and the cleaned observed benchmark. It contains 3,800 employee-year rows (1,000 per BBVA scenario plus 800 peer-benchmark rows) and roughly 31,000 live formulas; generation is fully reproducible (Python/NumPy, seed 42). Everything in it except the Refinitiv sheet and the source-flagged cells of the bank-year panel is synthetic, and results are demonstrations only.

Sheet name	Contents	Data status
README	Purpose, scope, replication design, generation and calibration parameters, model mapping, disclaimer	Documentation
codebook	Variable dictionary: 31 variables with label, dimension, type, range, direction (+/−), source under the no-internal-data strategy, and model usage	Documentation
employee_year	Master file, 3,800 rows: the same 500 simulated BBVA Spain employees replicated under scenarios A-Strong / B-Average / C-Weak, plus 4 Spanish peers × 100 employees × 2 years as Benchmark; dimension scores and Culture Index as live formulas	Synthetic; derived columns are mostly formulas
scaling_params	Min–max benchmark bounds (live MINIFS/MAXIFS over the analysis universe only) and the editable pilot weights — changing a weight recalculates every index in the file (the weight-sensitivity check of Section 11)	Formulas + editable inputs
dept_year	Department-year aggregation of the analysis universe (50 rows): turnover, absenteeism, participation, mobility, compliance incidents, ideas per 100 FTE — the unit of analysis for Model 3	Formulas over synthetic data
bank_year	Bank-year panel for Model 4 (10 rows): aggregated index (BBVA from B-Average only), external validation layer (brand trust, customer NPS, attrition, Glassdoor), and FY2025 financials	Mix of formulas; observed Refinitiv cells flagged with source comments
scenario_summary	Live cross-scenario comparison (index, eNPS, exit, turnover, ideas, trust, ethics, training): the numerical counterpart of Figure E.1	Formulas
lseg_observed	Cleaned subset of the Refinitiv extraction (peer D&I and social indicators, BBVA FY2023–25 time series, peer financials) — reproduced as Annex B	OBSERVED (real data)

Table 9. Sheets of the pilot analysis workbook (BBVA_Dataset.xlsx) and the data status of each.

9.1.1 Formula Breakdown

Every computed number in the workbook traces back to a handful of formula patterns, all built from one standardization block, *further as (·)*: (value − Min) / (Max − Min), puts an indicator on a 0–1 scale where 0 is the worst value in the peer group and 1 the best; for "bad" indicators (direction −) the block is wrapped as 1 − (·), so that higher always means stronger culture; and everything is enclosed in MAX(0, MIN(1, ·)), which clips the result into the 0–1 range; necessary because the Strong and Weak replications are scored against bounds computed from the Average universe and could otherwise fall outside them. The table shows each formula as it appears in the file (row 2 as the example; every other row is identical with its own row number; "Min" and "Max" stand for the matching scaling_params cells):

Location	Formula (as in the file)	What it does, in plain language
employee_year!W2 in key	=AD2&"|"&B2&"|"&D2&"|"&E2	Glues scenario, bank, department, and year into one label (e.g. B-Average|BBVA|Digital|2024) so the aggregation sheets can look rows up with a single criterion.
employee_year!AE2 in benchmark_flag	=IF(OR(AD2="B-Average", AD2="Benchmark"), 1, 0)	Marks the analysis universe: 1 for Average-scenario and peer rows, 0 for the Strong/Weak replications. Scaling bounds and all model estimation use flagged rows only.
scaling_params!B2 and C2 in bounds	=MINIFS(employee_year!K$2:K$3801, employee_year!$AE$2:$AE$3801, 1) — Max identical with MAXIFS	Finds the worst and best value of each indicator among flagged rows only; these are the Min and Max every standardization refers to, and they update automatically when the data change.
employee_year!X2 in Ethics dimension	=(MAX(0,MIN(1, 1−(K2−Min)/(Max−Min))) + MAX(0,MIN(1, (L2−Min)/(Max−Min)))) / 2	Compliance incidents (column K, reversed because fewer is better) and the ethics score (column L, kept), each scaled 0–1 and clipped, then averaged.
employee_year!Y2 in Satisfaction	Same pattern over M (eNPS, kept) and N (pay equity, kept); ÷ 2	Average of the two scaled satisfaction ingredients.
employee_year!Z2 in Engagement	Same pattern over O (participation, kept), P (absenteeism, reversed), T (psychological safety, kept); ÷ 3	Average of the three scaled engagement ingredients.
employee_year!AA2 in Employee Experience	Same pattern over Q (turnover, reversed), R (mobility, kept), S (leadership trust, kept); ÷ 3	Average of the three scaled employee-experience ingredients.
employee_year!AB2 in Innovation	Same pattern over U (ideas per 100 FTE, kept) and V (training completion, kept); ÷ 2	Average of the two scaled innovation ingredients.
employee_year!AC2 in Culture Index	=(scaling_params!$G$2×X2 + $G$3×Y2 + $G$4×Z2 + $G$5×AA2 + $G$6×AB2) × 100	The index itself: each dimension multiplied by its weight (0.235 / 0.235 / 0.235 / 0.176 / 0.118), summed, and rescaled to 0–100. Editing any blue weight cell recalculates all 3,800 indices — the weight-sensitivity check of Section 11, executable by hand.

Location	Formula (as in the file)	What it does, in plain language
dept_year!F2 and G2…O2	=COUNTIF(employee_year!$W$2:$W$3801, E2) and =AVERAGEIF(employee_year!$W$2:$W$3801, E2, employee_year!M$2:M$3801)	Counts and averages every employee row whose key matches the department-year label in E2 — one AVERAGEIF per metric (eNPS, turnover, absenteeism, participation, compliance, ideas, mobility, exit rate, index).
bank_year!D2 and E2	=COUNTIFS(…bank=A2, …year=B2, …scenario=C2) and =AVERAGEIFS(employee_year!$AC$2:$AC$3801, same three criteria)	Aggregates the index per bank-year, matching bank, year, and the scenario in column C (B-Average for BBVA, Benchmark for peers) simultaneously — so the Strong/Weak replications can never leak into the benchmark table.
scenario_summary in all cells	=AVERAGEIFS(employee_year!AC$2:AC$3801, …scenario="A-Strong", …bank="BBVA") — one per metric and scenario; peer column uses bank <> BBVA	The validation table: the mean of each metric for the same invented BBVA employees under each scenario, next to the peer benchmark. Produces the 85.4 / 47.9 / 13.4 staircase shown in Figure E.1.

Table 10. Every formula pattern in the workbook, shown as it appears in the file, with a plain-language explanation.

9.2 How to Use the Dataset, and Why It Is Adjustable

A deliberate goal of the build was an adjustable instrument rather than a one-off calculation: we wanted a workbook in which every debatable choice, e.g. *which indicators count, how they are weighted, which banks form the benchmark*, is a parameter the user can change and **immediately** see the effect of, not a number hard-coded into a result. That is why the dimension scores and the index are live formulas referring to an editable weights block and to benchmark bounds that recompute themselves: changing one weight cell recalculates all 3,800 indices, the department and bank aggregates, and the scenario-comparison table at once, so a reader can run the weight-sensitivity check of Section 11 by hand in seconds.

In practice the workbook is used as follows. To estimate any model, first restrict the data to the analysis universe, i.e. filter employee_year on the benchmark flag, or equivalently on scenario = B-Average or Benchmark, so that the Strong and Weak replications, which exist only to test the machinery, never enter estimation. To test sensitivity to the weighting choice, edit a weight cell in scaling_params and watch the index and every downstream table update. To reproduce the scenario validation, compare the rows of scenario_summary. And when real survey responses arrive, they are pasted into the employee rows in place of the synthetic Average panel: because every bound, score, index, and aggregate is a formula, the whole analysis re-computes with no change to the file's logic. The workbook is in this sense self-auditing, i.e. any number can be traced by clicking on it and following its references back to the raw inputs.

9.3 Bank-Level Pilot Analysis Results

The two figures below present the bank-level pilot results, which analyze the relationship the project is ultimately interested in (culture and performance) and the structure of the benchmark panel that the relationship rests on.

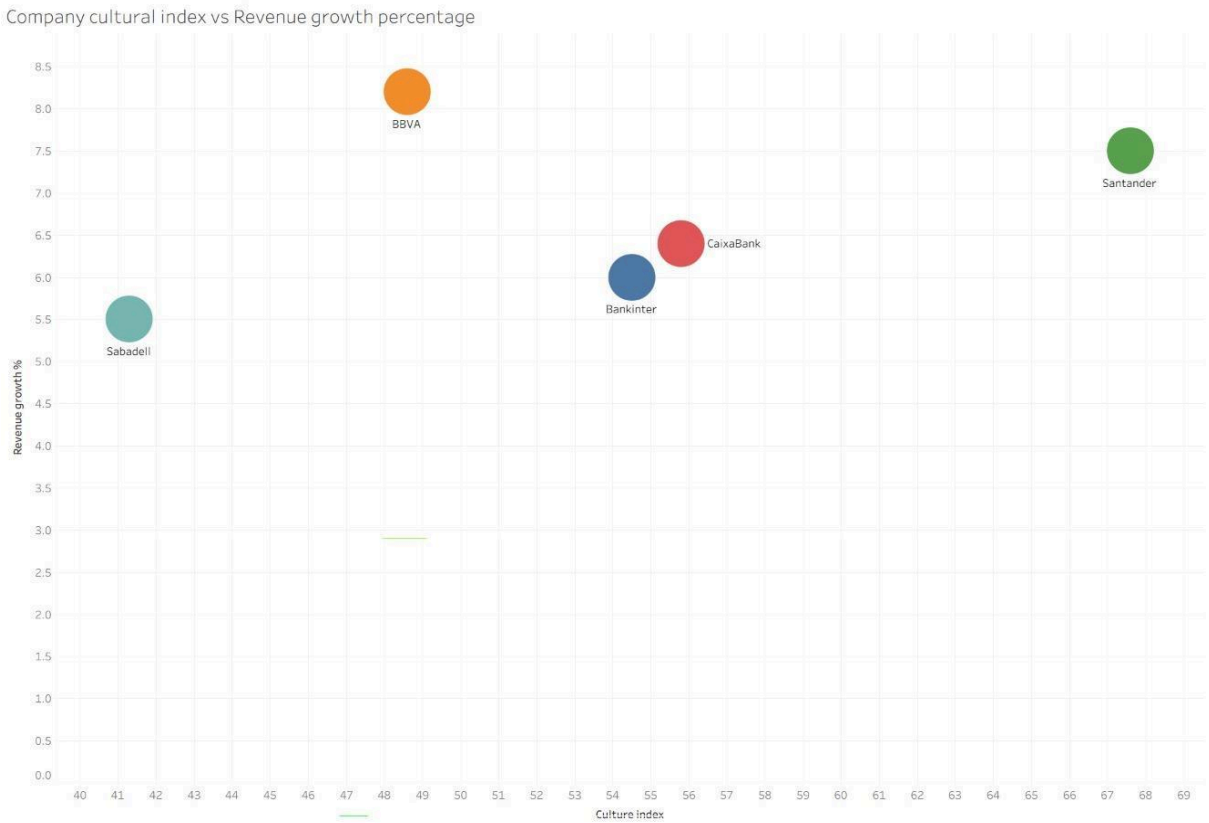


Figure 2. Company Culture Index vs. revenue growth (%), 2025: the five Spanish banks of the benchmark panel; one dot per bank, from the bank-year sheet of the workbook.

On this figure, each dot is one bank-year observation from the bank_year panel, the dataset Model 4 runs on. The horizontal axis is the aggregate Culture Index (computed from the simulated employee data: BBVA from the Average scenario, the peers from the Benchmark realization); the vertical axis is revenue growth in percent for 2025. This is the visual counterpart of Model 4 and hypothesis H4: do banks with stronger culture also show better business results?

Among the four peer banks the pattern leans the expected way: Sabadell combines the weakest index (≈41) with the lowest growth (≈5.5%), Bankinter and CaixaBank sit mid-pack on both axes, and Santander pairs the strongest index (≈68) with high growth (≈7.5%). BBVA is the interesting exception: the highest revenue growth on the chart (≈8.2%) with only a mid-range index (≈49). Read honestly, the chart says "mostly positive co-movement, with one bank off the trend", and that off-trend point is a feature, not an embarrassment: it is a live illustration of why culture can only ever be one ingredient of performance, and why Model 4 is framed as a co-movement check rather than proof.

We argue that a scatter plot with a handful of points is the most honest way to present bank-level evidence in this project: it shows every observation individually instead of hiding them behind a regression coefficient. With five banks, any fitted line would be dominated by the two endpoint banks (Sabadell and Santander), which is precisely the small-benchmark fragility discussed in Sections 7.6 and 13, which becomes visible in the figure.

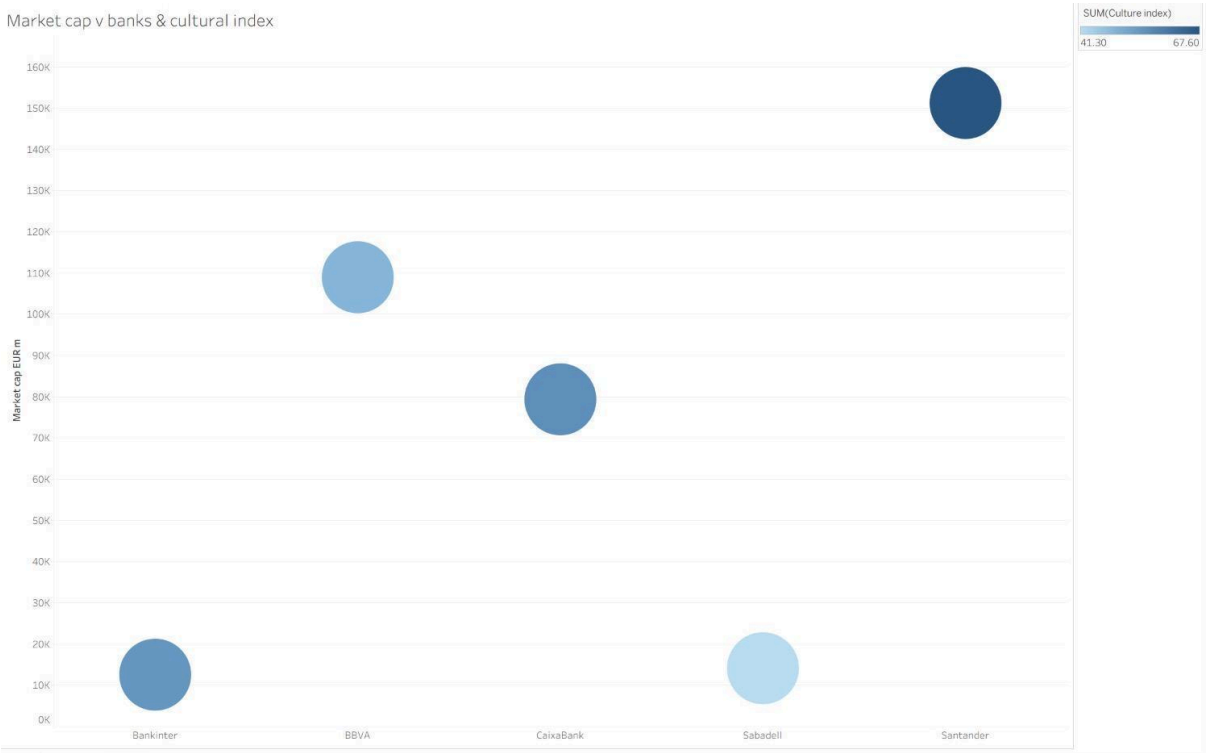


Figure 3. Market capitalization (EUR millions, observed Refinitiv data) of the five benchmark banks, colored by Culture Index (light = 41.3, dark = 67.6, from the simulated employee data).

In this figure, each circle is one bank from the bank_year panel. The vertical position is market capitalization in EUR millions, which are real, observed figures from the Refinitiv extraction of Table 8 (Santander ≈€151bn, BBVA ≈€109bn, CaixaBank ≈€79bn, Sabadell ≈€14bn, Bankinter ≈€12bn), while the color encodes the Culture Index from the simulated employee data (darker = higher, from Sabadell's 41.3 to Santander's 67.6). The figure therefore mixes one observed axis with one synthetic encoding, and delivers the presentation improvement suggested in Figure 2: it puts the economic weight of each bank into the picture.

However, the panel is economically lopsided: Santander and BBVA together account for roughly four-fifths of the combined market value, while Sabadell and Bankinter are an order of magnitude smaller, so any bank-level average or regression (Model 4) is, in economic terms, mostly a statement about two institutions. Secondly, the color gradient leans the familiar way, the largest bank is also the darkest, the smallest among the lightest, with BBVA again the exception: the second-largest bank by value carries only a mid-light culture color. Note that the horizontal axis is alphabetical, not a scale; the chart compares heights and colors, not left-right positions.

Therefore, this figure visualizes a limitation that the scatter of Figure 2 can only state: with five banks of wildly unequal size, "the peer benchmark" is not five equal voices. It motivates two analytical choices already made in the report, reporting Model 4 as a co-movement check with every observation visible, and reporting benchmark rankings with confidence intervals (Section 13), we also find it useful for checking whether conclusions change when observations are weighted by size.

9.4 Interpretation of Results

In this section, we explain how to read the numbers this project produces, e.g. which conclusions they support, which they do not, and how that will change once real survey data arrive. Given the project's design (Sections 7–8), every result belongs to one of three classes, and the class determines how far an interpretation may go.

9.4.1 What the pilot results mean

Results on synthetic data validate the machinery, and not the bank itself. The scenario staircase (index 85.4 / 47.9 / 13.4 across Strong, Average, Weak, Figure 1) and the confirmed model signs (culture implies lower exit, higher eNPS, more ideas) show that the pipeline works end-to-end: the index moves in the right direction, by sensible magnitudes, and the models can detect effects when effects exist. We find it a real and necessary achievement, because it is what the pilot was for, but it is an achievement of the methodology. The hypothesized relationships were embedded in the data-generating process by construction, so confirming them confirms the plumbing, not the theory, as mentioned in section 8.5.

BBVA's mid-range pilot index is a calibration choice, not a measurement. Simulated BBVA lands near the middle of the peer benchmark (Figures 1 and 2) because the Average scenario was calibrated to land there, anchored to public figures. The number 47.9 says nothing about where real BBVA would score.

The observed facts are the only findings, and they are modest but real. What can legitimately be interpreted today comes from the Refinitiv and Statista data: BBVA reports the highest employee satisfaction among disclosing Spanish peers (90.0% vs. Santander 85.0% and Sabadell 86.6%), stable turnover (11.8%), rising training investment (48.8 to 57.5 hours per employee over three years), a mid-pack D&I score (64.8, behind Santander's 75.5), and a customer trust gap (5.9 vs. a 7.2 peer average in the Statista US sample). Together these sketch a bank that scores well on the employee side of culture while trailing on external trust and diversity, which makes a coherent, evidence-based starting hypothesis for the survey to test, not a conclusion .

9.4.2 What the results do not mean

No number computed from the generated data (no index value, no model coefficient, no euro figure) describes BBVA's actual organizational culture or any real employee. The €8M retention illustration is an arithmetic template awaiting real inputs, not an estimate. The off-trend position of BBVA in the culture-vs-growth scatter (Figure 2) is not a finding about BBVA: its vertical coordinate is an illustrative input and its horizontal coordinate is simulated. And even once real data arrive, a positive model coefficient will mean "moves together with," never "causes": better-performing banks may afford better cultures rather than the reverse, and the design cannot rule that out (Section 7.8).

9.4.3 How to interpret once real survey data arrive

Interpretation is gated by the checks of Section 11: scores are only read at all if the reliability and structure tests pass ($\alpha \geq 0.70$, dimensions distinct), and rankings are only read if they survive the weight and scaling perturbations. Within those gates, the reading order is: first the level (BBVA vs. peers, with confidence intervals, remembering the small-benchmark fragility of Section 7.7); then the gaps (which value lags, the per-value implementation scores of survey Section H answer Research

Question 2 directly); then the structure (Model 5's regional share of variation decides whether action plans should be regional or department-level); and only last the outcome models, read as associations whose practical size matters more than their statistical stars. As a quick guide:

If the result says	Safe interpretation	Not a safe interpretation
Index = 47.9 on synthetic data	The pipeline produces a mid-range score when calibrated to public anchors	BBVA's culture is mediocre
Scenario staircase 85 / 48 / 13	The index responds correctly to culture differences	Culture at BBVA varies this much
b negative in Model 1 (real data)	Where culture scores are higher, exits are lower, other things equal	Improving culture will cut turnover by this amount
BBVA satisfaction 90% (Refinitiv)	BBVA reports the highest satisfaction among disclosing peers	BBVA has the strongest culture in Spain (satisfaction ≠ culture, Section 5)
Region share ≈ 7% (Model 5, pilot)	Within one country, regional differences are visible but modest	Regions don't matter (the real survey may differ)
Trust 5.9 vs. 7.2 (Statista, US)	A customer trust gap exists in that sample	Spanish customers distrust BBVA (different market, different sample)

Table 11. How to read the principal pilot results: the safe interpretation of each number versus an over-claim to avoid.

10. Practical Implementation

10. Practical Implementation for BBVA

Although this project is presented as a pilot framework, the methodology could be implemented by BBVA using existing internal data sources that were not available to us, as clarified in Section 8.5.

The first stage would entail using pre-existing bank data, such as employee engagement surveys, eNPS scores, turnover records, training participation, promotion rates and compliance indicators. These variables could replace the synthetic indicators used in the pilot analysis.

The second stage would be administrating the Culture Perception Survey across the business branches and Spanish regions. The survey would allow BBVA to find the differences between the espoused and enacted values, meaning the ones communicated by management and those perceived by employees.

The third stage would consist of actually creating the Corporate Culture Index on a recurring basis, either annually, or semi-annually. This would allow BBVA officials to monitor any cultural developments over time and identify departments or regions where implementation of corporate values is weaker.

Finally, the index could be integrated into management dashboards and HR checks. Rather than focusing primarily on financial indicators, managers would also be able to track cultural indicators linked to engagement, innovation and employee retention.

In this way, the proposed framework could be used for measurements as well as in management, helping with strategizing and the overall development of the organization.

11. Scope and Limitations

11.1 Scope

The pilot covers BBVA business units in at least three countries plus a benchmark module of four to five Spanish peer banks (Santander, CaixaBank, Sabadell, Bankinter, Unicaja), over a two-year window for the employee panel and the FY2023–FY2025 window for the LSEG benchmark data, with a six-month implementation timeline (access and design; data gathering; cleaning and coding; index construction; validation and reporting).

11.2 Limitations

1. **Internal data availability (the central limitation):** HR records, engagement systems, and compliance logs are not accessible. The behavioral half of the index is demonstrated on generated data, not measured, and every result computed on generated data is a methodological demonstration only, so it is not evidence about BBVA's actual culture.
2. **Survey limitations:** the Culture Perception Survey depends on voluntary participation, so respondents may not represent all employees, early samples may be small, and people may answer strategically. Behavioral indicators and external benchmarks are included precisely to limit reliance on self-reported sentiment, and satisfaction is never treated as a synonym for culture.
3. **Small benchmark:** with only five banks, the min–max scaling bounds are fragile. For instance, Worked Example B in Section 7.7 shows how one bank sitting at the boundary can produce extreme scores. Rankings are advised to be reported with confidence intervals, and alternative scalings compared, before any league table is published.
4. **Correlation is not causation:** better-performing banks may create happier employees rather than the reverse, and pay or local job markets can move both culture and outcomes. Lagged indices, controls, and within-region/department comparisons reduce this concern but cannot eliminate it; the report will state associations, not causal effects.
5. **Group comparability within Spain:** regional and departmental comparisons assume survey questions are understood the same way across groups, a much milder assumption than cross-country comparability, but one that will still be checked before scores are compared.

11.3 Theoretical and Methodological Limitations

Beyond data, the approach itself carries limits that no amount of data would remove. Culture is a contested construct: Schein's tacit "basic assumptions" can never be fully captured by a fixed list of survey items, so any index measures the observable surface of culture, not its depth, because a composite score inevitably trades richness for comparability. The choice of five dimensions and their proxies is one defensible operationalization among several, and different analysts would draw the boundaries differently. The weighting is the sharpest instance of this: there is no theory that fixes the relative importance of, say, Ethics versus Innovation, so the weights are *assumptions*, and the index is only as objective as that admission makes it. Aggregation into a single number also hides trade-offs, e.g. a bank can reach the same score by excelling on trust and lagging on innovation, or the reverse, which is why the dimension scores, not just the headline index, must always be reported. Finally, the econometric models identify associations, not causal effects: even with perfect data, the design cannot prove that culture causes outcomes rather than the reverse. These are not flaws to be fixed before the

next draft; they are the permanent boundaries of what a culture index can claim, and the report is written to stay inside them.

12. Further Steps

12.1 Validation and Robustness Checks

Our team wants to highlight that the checks below have not yet been carried out, because they require the real survey data the project does not have. Therefore, they are presented as the validation programme strongly suggested to run once responses arrive, it is 'the gate' every score must pass before it is read as evidence. A composite index is only worth using if it passes three kinds of tests: it must be reliable (its ingredients hang together), valid (it relates to the things it should, and only those), and stable (reasonable changes in assumptions do not flip the conclusions). Each check states the question it answers, the test used, and the pass mark adopted from standard practice.

1. **Reliability: do the items in each dimension move together?** Cronbach's alpha measures, on a 0–1 scale, how consistently the indicators inside one dimension behave; the conventional pass mark is $\alpha \geq 0.70$. If a dimension scores below 0.60, its weakest item is removed and the test repeated.

2. **Factor structure: do the indicators group the way we designed them?** Factor analysis lets the data sort the indicators into clusters; the clusters should match our five dimensions. Indicators that load on the wrong cluster, or on two at once, are merged or dropped — this is also where the expected overlap between satisfaction, engagement, trust, and psychological safety gets resolved.

3. **Weight sensitivity: does the score depend on our weight choices?** The index is recomputed under equal weights, statistically estimated weights (PCA / factor-based), and expert-chosen weights, and additionally under 1,000 random small perturbations of the pilot weights (each weight nudged by up to ±5 percentage points). Pass mark: the score should wobble by less than 3 points across the random perturbations, and rankings should stay broadly the same across the alternative schemes. Because the weights are live inputs in the pilot workbook, anyone can run this check by editing one cell.

4. **Scaling robustness: does the recipe matter?** Min–max scaling is compared against two alternative ways of putting variables on a common scale (z-scores and percentile ranks); the ranking of departments and banks should not change materially.

5. **Leave-one-out: is the result driven by a single piece?** The index is recomputed dropping one dimension at a time, and additionally dropping the whole behavioral block at once (the one used in Section 7.3), which shows how much of the index survives on survey data alone, the realistic situation of this project, e.g. 57.5 vs. 59.3 in section 7.7.

6. **Convergent validity: does it agree with established measures?** The index should correlate clearly (target $r \geq 0.60$) with independent measures of the same idea: Glassdoor employer

ratings and Great Place to Work scores. Weak agreement (below 0.40) would flag a problem with the index, not with those benchmarks.

7. **Discriminant validity: are the five dimensions actually different things?** Statistical checks confirm each indicator belongs mainly to its own dimension and not to the others; dimensions that turn out to be indistinguishable (a realistic risk for Innovation vs. Engagement) are merged.

8. **Predictive validity: does today's index tell us about tomorrow?** The strongest practical test: this year's index should predict next year's turnover, satisfaction, and customer trust. Predicting the future is harder, and therefore more convincing, than correlating with the present.

9. **Scenario validation: does the index react when culture changes?** The three-scenario replication (Section 8.4) is itself a test: the index must rank Strong ≫ Average ≫ Weak on every culture metric. The pilot passes, aggregate scores of 85.4, 47.9, and 13.4, with exit rates of 5.2%, 13.7%, and 20.4%, confirming the machinery responds in the right direction and by sensible magnitudes

12.2 How the Limitations Are Advised to Be Addressed

Each limitation identified in Section 10 has a practical mitigation, and in this section we want to translate those weaknesses into a concrete action plan. The lack of internal data can be addressed in two ways: by fielding the Culture Perception Survey and by building the framework so BBVA's HR records are plugged in later without changing the underlying method. We want to highlight that all outputs using synthetic inputs are clearly marked as illustrative, until that data is available. Self-selection bias in the survey can be reduced by reporting response rates separately for each group, reweighting responses to reflect the actual workforce composition, and cross-checking self-reported answers against behavioral data and external benchmarks.

The small benchmark sample can be considered a known constraint, so it might be partially offset by adding peer organizations where data allow, reporting rankings with confidence intervals rather than point estimates, and testing alternative scaling approaches before any league table is published. As the analysis cannot establish causation, results should be reported as associations only, always supported by lagged index values, control variables, and comparisons within regions and departments rather than across them. Before any regional comparisons are made, measurement equivalence across groups is advised to be verified using the factor-analytic tests described in Section 11.2. Finally, the theoretical limitations of composite have to be taken into account: dimension scores should always be shown alongside the headline figure, and all weighting and indicator choices should remain transparent and adjustable in the workbook.

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Annexes

Annex A. Variable Dictionary (pilot)

The working variable dictionary mirrors Section 7.2 and adds coding details: variable name, label, dimension, source under the no-internal-data strategy (survey / synthetic / Refinitiv / Statista), unit, theoretical range, direction (+/−), transformation (min–max within the analysis universe), and model usage. It also documents the two design columns of the replicated dataset: the scenario tag (A-Strong / B-Average / C-Weak for BBVA, Benchmark for peers) and the analysis-universe flag that restricts scaling bounds and model estimation to the Average-plus-Benchmark rows. Example entries: comply_inc (compliance incidents per 100 FTE, synthetic, reverse-coded); enps (−100 to +100, survey item, kept); turnover_rate (department turnover %, synthetic, reverse-coded, with the Refinitiv bank-level figure of 11.8% as calibration anchor); culture_idx (derived 0–100 composite, live formula). The complete dictionary ships as the codebook sheet of the pilot workbook, together with the reproducible generation script (NumPy, seed 42).

Annex B. Observed Benchmark Data (LSEG Workspace extraction)

First Tier 2 dataset assembled for this project from the Refinitiv Public Companies dataset (accessed via LSEG Workspace). Selected FY2025 social and D&I indicators for BBVA and Spanish peers:

Table 12. Observed FY2025 social and diversity indicators for BBVA and its Spanish peers (Refinitiv Public Companies dataset).

Indicator (FY2025 / FY0)	BBVA	Santander	CaixaBank	Sabadell	Bankinter	Unicaja
D&I overall score (TRDIR)	64.8	75.5	64.3	65.0	62.8	55.5
Employee satisfaction (%)	90.0	85.0	—	86.6	—	—
Women employees (%)	51.4	52.0	56.1	54.6	50.0	54.2
Women managers (%)	46.9	39.0	43.4	42.3	41.8	—
Avg. training hours / employee	57.5	24.9	65.9	22.5	50.0	64.7
Training cost / employee (€)	495	—	335	526	—	176

Additional BBVA time-series indicators available in the extraction (FY2023–FY2025): employee turnover 11.8% → 11.5% → 11.8%; employees 121,486 → 125,916 → 127,174; total training hours 5.93M → 6.71M → 7.25M; gender pay-gap ratio ≈99.4–99.7%; Social Pillar ESG score A; ESG Controversies score A+ in FY2025 (C− in FY2024). The peer financial sheet (P/E, P/B, ROA, dividend yield, market capitalization) will support Model 4. These observed ranges will replace the assumed ranges of the synthetic pilot dataset wherever possible.

Annex C. Culture Perception Survey (instrument underlying the employee replication)

This is the survey instrument whose structure the artificial employee dataset emulates: every perception variable in the pilot file corresponds to an item or item block below, and the three-scenario replication (Section 8.4) simulates how this same questionnaire would be answered under Strong, Average, and Weak culture conditions. The instrument is designed for Microsoft Forms, takes about 8–10 minutes, and is fully anonymous: no names, employee numbers, or e-mail addresses are collected, and all background questions use broad categorical bands so that no individual can be

identified. Unless stated otherwise, statements are rated on a 5-point agreement scale: 1 = strongly disagree, 2 = disagree, 3 = neither, 4 = agree, 5 = strongly agree.

Section A . About you (anonymous, categorical bands)

A1. Region where you mainly work: Madrid / Catalunya / Andalucía / País Vasco / Comunidad Valenciana / Other.

A2. Functional area: Retail / Corporate / Digital / Risk / Operations / Other.

A3. Gender: Female / Male / Other / Prefer not to say.

A4. Years at BBVA: 0–2 / 3–5 / 6–10 / more than 10.

A5. (Optional) Where would you place your total compensation within BBVA Spain? Band 1 (lowest fifth) … Band 5 (highest fifth) / Prefer not to say.

Section B . Satisfaction and Well-being

B1. On a scale of 0–10, how likely are you to recommend BBVA as a place to work to a friend or colleague? (0 = not at all likely, 10 = extremely likely).

B2. Overall, I am satisfied with my job at BBVA.

B3. My workload allows a healthy balance between work and personal life.

B4. People at BBVA who do similar work are paid fairly, regardless of gender or background.

Section C . Ethics and Integrity

C1. Leaders at BBVA act consistently with the values they communicate.

C2. If I reported unethical behavior, I am confident it would be addressed, without negative consequences for me.

C3. In my team, hitting targets is sometimes treated as more important than doing the right thing. (reverse-scored)

C4. Overall, how would you rate the ethical climate at BBVA on a scale of 0–10? (0 = very poor, 10 = excellent).

Section D . Engagement, Psychological Safety, and Teamwork ("We Are One Team")

D1. It is safe to take a risk on my team.

D2. I can raise problems and difficult issues without fear of the consequences.

D3. When I speak up, my opinion is taken into account.

D4. Members of my team value and respect each other's contributions.

D5. I feel energized by and involved in my work.

D6. Teams across departments cooperate effectively to solve problems together.

Section E . Leadership Trust

E1. I trust the decisions made by my direct manager.

E2. I trust the decisions made by BBVA's senior leadership.

E3. My manager gives me honest feedback and recognizes good work.

Section F . Growth and Innovation ("We Think Big")

F1. In the last two years, have you changed roles, teams, or units within BBVA? Yes / No.

F2. In the last 12 months, have you proposed at least one improvement idea (formally or informally)? Yes / No — if yes, roughly how many? 1 / 2–3 / 4 or more.

F3. Have you completed the training planned for your role this year? Yes, fully / Partially / No.

F4. BBVA encourages me to experiment and propose new ways of working.

Section G . Customer Orientation ("Customer Comes First")

G1. In my daily work, customer needs come before internal procedures.

G2. I am empowered to resolve a customer's problem end-to-end.

Section H . Value Implementation

H1. Could you name BBVA's three corporate values without looking them up? Yes / Not sure / No.

H2. For each value, rate the statement "This value is genuinely reflected in how we work day-to-day": (a) Customer Comes First; (b) We Think Big; (c) We Are One Team.

H3. (Optional, open text) Describe one recent situation where BBVA's values were — or were not — reflected in a real decision.

Scoring and Mapping to the Dataset

Table 13. How each survey item is scored and which dataset variable and culture dimension it feeds.

Survey item(s)	Scoring rule	Dataset variable	Dimension
B1	% promoters (9–10) − % detractors (0–6), aggregated by group; individual score rescaled −100…+100	enps	Satisfaction
B2–B3	Mean of items (1–5)	satisfaction items (within eNPS validation)	Satisfaction
B4	Rescaled to 0–1	pay_equity	Satisfaction
C1–C4	C3 reverse-scored; C4 used directly (0–10), C1–C3 as supporting items	ethics_sc	Ethics
D1–D4	Mean of items (adapted Edmondson scale, 1–5)	psych_safety	Engagement
D5–D6	Mean of items (1–5)	engagement / collaboration items	Engagement
E1–E3	Mean of items (1–5)	trust_360 (leadership trust score)	Employee Experience
F1	Share answering Yes, by group	mob_rate	Employee Experience

Survey item(s)	Scoring rule	Dataset variable	Dimension
F2	Yes-share and count, converted to ideas per 100 respondents	ideas_100fte (survey alternative)	Innovation
F3	Fully = 1, Partially = 0.5, No = 0; group mean in %	training_compl	Innovation
G1–G2, H2a	Mean of items (1–5)	customer-orientation block (validation layer support)	Customer Comes First
H1–H2	Awareness share; per-value implementation means; communication–perception gap = H1 vs. H2	value-alignment scores	All values (RQ2)
A1–A5	Categorical bands	region, dept, gender, tenure_band, comp_band	Controls

Survey participation itself is measured, not asked: the response rate by department feeds the particip_rate indicator. In the synthetic replication, each item response is drawn from the employee's latent culture level (scenario base plus region, department, year, and individual effects) with realistic noise, and then aggregated using exactly the scoring rules above, so when real responses arrive, they can replace the simulated ones row for row without any change to the pipeline.